

# Project 1 Report: ROS 2 and MoveIt Setup with Panda Robot Simulation

## Student Details

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**Project Title:** ROS 2 Jazzy Installation and Panda Robot Motion Planning Simulation

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## Objective

The objective of this project was to install ROS 2 Jazzy, configure the required robotics packages, and successfully launch a robotic arm simulation using MoveIt and RViz. The project focused on understanding robot motion planning, inverse kinematics, and basic robot interaction in a simulated environment.

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## Software and Tools Used

- Ubuntu 24.04 LTS (WSL2)
  - ROS 2 Jazzy
  - MoveIt 2
  - RViz2
  - Panda Robot (Franka Emika Panda)
  - ROS 2 Control
  - Ubuntu Terminal
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## Installation Steps

1. Installed Ubuntu 24.04 on WSL2.
2. Updated system packages.
3. Added ROS 2 package repositories.
4. Installed ROS 2 Jazzy Desktop.
5. Configured ROS environment variables.
6. Installed MoveIt 2 packages.

7. Installed ROS2 Control CLI tools.

8. Verified ROS installation using:

- `ros2 --help`
  - `rviz2`
  - `ros2 node list`
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## Simulation Setup

The Panda robot simulation was launched using MoveIt.

### Launch Command

```
ros2 launch moveit_resources_panda_moveit_config demo.launch.py
```

The simulation successfully started and opened RViz with the Panda robotic arm model.

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## Tasks Performed

### 1. Robot Visualization

- Loaded Panda robot in RViz.
- Verified robot links and joints.
- Confirmed robot description and semantic configuration were loaded.

### 2. Motion Planning

- Used MoveIt Motion Planning Plugin.
- Selected the planning group `panda_arm`.
- Manipulated the end-effector using interactive markers.
- Generated valid trajectories using the Plan button.

### 3. Joint Manipulation

- Modified joint values through the Joints tab.
- Observed forward kinematics behavior.
- Verified robot pose updates in real time.

### 4. Scene Object Creation

- Added collision objects (Box).
- Positioned objects in the workspace.
- Verified scene updates.

## 5. Controller Verification

Executed:

```
ros2 control list_controllers
```

Result:

- panda\_hand\_controller detected and active.

Executed:

```
ros2 control list_hardware_interfaces
```

Result:

- All Panda arm joints were available.
  - Hardware interfaces loaded successfully.
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## Results

The Panda robotic arm was successfully visualized and controlled inside RViz.

Achievements:

- ROS 2 Jazzy installed successfully.
  - MoveIt 2 configured successfully.
  - Panda robot loaded successfully.
  - Motion planning executed successfully.
  - Joint manipulation tested successfully.
  - Collision object creation tested successfully.
  - Controller interfaces verified successfully.
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## Learning Outcomes

Through this project, the following concepts were learned:

- ROS 2 architecture
- Nodes and topics
- Robot Description (URDF)
- Motion Planning
- Forward Kinematics
- Inverse Kinematics
- RViz Visualization

- MoveIt Framework
  - Robot Controllers
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## Conclusion

The project successfully demonstrated the setup and operation of a robotic arm simulation using ROS 2 Jazzy and MoveIt 2. The Panda robot was visualized, manipulated, and motion-planned in RViz. This project provided a strong foundation for future robotics projects involving navigation, SLAM, autonomous robots, and robotic manipulation.